A First-Return Fredholm Family for the Thue–Morse Renewal Shift

Route-A structural certificate C164 — internal round 0

Abstract

Let S contain the indices at which the Thue–Morse bit is one and let $F(z) = \sum_{s\in S} z^{s+1}$ be the first-return series of the associated renewal shift. We retain each return word as a Hilbert coordinate and construct a rank-one family K_z which is holomorphic in trace norm on the open unit disk. Every trace power is exact, $\operatorname{Tr} K_z^m = F(z)^m$, and adjoining the all-zero fixed orbit gives $\det_F(I - ([z] \oplus K_z)) = (1-z)(1-F(z))$, the inverse source zeta proved in C159. This preliminary certificate already separates return-branch ownership from a scalar determinant chosen after the fact. It does not identify the induced family with the time-one adjacency, nor claim a target divisor, arithmetic factorization, self-adjoint lift, or Hilbert–Pólya operator.

Keywords: symbolic dynamics; Thue–Morse sequence; renewal shift; first-return operator; trace class; Fredholm determinant.

中文摘要

令集合 S 由 Thue–Morse 序列中取值为一的下标组成， $F(z) = \sum_{s\in S} z^{s+1}$ 为相应更新移位的首返级数。本文不把 F 压缩成事后选定的 一维标量，而以每条返回字为希尔伯特坐标，构造单位圆盘内迹范数全纯的秩一算子族。它的全部迹幂均精确等于 $F(z)^m$；再加入全零不动轨道，所得 Fredholm 行列式恰为 C159 源动力系统的逆 zeta。此初稿只建立首返算子所有权，不把它混同于单步邻接， 也不声称目标除子、算术分解或自伴算子实现。
关键词：符号动力系统；Thue–Morse 序列； 更新移位；首返算子；迹类；Fredholm 行列式。

1 Frozen source and branch construction

Write $t_s \in \{0,1\}$ for binary digit-sum parity and $S = \{s \geq 0 : t_s = 1\}$. The renewal code is $\mathcal{C} = \{10^s : s \in S\}$; one shift is one clock tick and a codeword 10^s returns after $s+1$ ticks. C159 proves

$$\zeta_{X_S}(z)^{-1} = (1-z)(1-F(z)), \qquad F(z) = \sum_{s\in S} z^{s+1}. \tag{1}$$

On $\mathcal{H} = \ell^2(S)$ freeze

$$q_s = e^{-\sqrt{s+1}}, \quad u = (q_s), \quad \ell_z(f) = \sum_{s\in S} q_s^{-1} z^{s+1} f_s, \quad K_z f = \ell_z(f) u. \tag{2}$$

The branch s itself is retained as $B_s(z)f = q_s^{-1} z^{s+1} f_s u$, so $K_z = \sum_s B_s(z)$ rather than a scalar written down from (1).

2 Trace ownership

The vector u is square summable. For every $\rho < 1$,

$$\sum_{s\in S} \|B_s(z)\|_1 \leq \|u\|_2 \sum_{s\in S} e^{\sqrt{s+1}} \rho^{s+1} < \infty \quad (|z| \leq \rho). \tag{3}$$

The differentiated series obeys the same comparison. Thus (2) is a trace-norm holomorphic rank-at-most-one family. Gauge factors cancel: $\operatorname{Tr} B_s(z) = z^{s+1}$ and $\ell_z(u) = F(z)$. Consequently

$$K_z^m = F(z)^{m-1} K_z, \quad \operatorname{Tr} K_z^m = F(z)^m, \quad \det_F(I - K_z) = 1 - F(z). \tag{4}$$

With $\mathcal{L}_z = [z] \oplus K_z$, (1) and (4) give

$$\det_F(I - \mathcal{L}_z) = (1-z)(1-F(z)) = \zeta_{X_S}(z)^{-1}. \tag{5}$$

The branch sum, compact-subdisk convergence, and all-power trace law are the ownership certificate. Equation (5) alone, realized by a post-hoc scalar, would be tautological.

3 Preliminary boundary audit

The uninduced one-step adjacency remains a different operator:

$$A\delta_n = \delta_{n+1} + t_n\delta_0. \tag{6}$$

This round-zero draft records only that the standard unweighted completion does not make its infinite return row bounded. A conclusion about all diagonal weighted Hilbert completions requires a separate quantified proof and is therefore not asserted in this snapshot.

The 128-bit, 32-branch, and degree-48 ledgers are finite sentinels. The strict tuple is

$$\begin{aligned}(&\texttt{A1\_WEAK}, \texttt{A2\_FAIL},\\ &\texttt{A3\_PARTIAL\_ANALYTIC\_STRUCTURE}, \texttt{A4\_FAIL}).\end{aligned}$$

We claim no target divisor or counting law, arithmetic/local factor, target functional equation, root number, automorphy, unitary/Hamiltonian or natural self-adjoint lift, Hilbert–Pólya construction, or Route-B authorization. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and code expose the producer, independent checker, symbolic reconstruction, replay, and mutation audit.

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Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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